WySGNN: Wyckoff-Position-Aware Space Group Equivariant Networks for Crystal Tensor Property Prediction

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Abstract

Equivariant graph neural networks enforce continuous Euclidean $E(3)$ symmetry but are usually described as agnostic to the *discrete* crystallographic site- and space-group symmetry that, by Neumann's principle, forces specific components of a physical tensor to vanish. We introduce WySGNN, which makes this symmetry architectural: after each message-passing layer every atom's features are projected onto its Wyckoff site-symmetry-invariant subspace, and the readout projects the predicted tensor onto the crystal-point-group-invariant subspace, yielding piezoelectric tensors that satisfy Neumann's principle exactly (e.g. identically zero for centrosymmetric crystals). Our central theoretical result is a clarifying one: for an *exactly* $E(3)$-equivariant network on a symmetric crystal, both projections are provably identities on the network output—continuous equivariance already implies the discrete symmetry. The projections are therefore valuable as (i) an exactness guarantee robust to numerical equivariance error and (ii) symmetry that the dominant *invariant-feature* property models (CGCNN/MEGNet-style) genuinely lack. On piezoelectric tensors we verify this empirically: equivariant variants are indistinguishable (Frobenius spread $0.00e+00$), all symmetry-aware models have exactly zero violation rate, while the invariant baseline violates symmetry at rate 0.979. Code and all numbers are released and reproducible from a single script.

1 Introduction

Many technologically central material properties are *tensors*: the piezoelectric tensor d_{ijk} (rank 3), the elastic tensor (rank 4), the dielectric tensor (rank 2). Crystallographic symmetry, through Neumann's principle Nye [1985], dictates that the crystal's point group must leave the property tensor invariant; this forces many components to vanish and ties others together. For instance every component of a rank-3 polar tensor vanishes in a centrosymmetric crystal.

$E(3)$-equivariant message-passing networks (NequIP Batzner et al. [2022], MACE Batatia et al. [2022], Allegro Musaelian et al. [2023], built on tensor field networks Thomas et al. [2018] and `e3nn` Geiger and Smidt [2022]) are equivariant under continuous rotations. A common motivating claim is that they do not encode the discrete crystallographic symmetry that governs which tensor components may be nonzero. We take this claim seriously, make the discrete symmetry architectural, and—guided by an implementation in which every component is unit-tested—arrive at a more precise picture.

Contributions.

- **Site-symmetry (Wyckoff) projection** as a per-atom architectural constraint: $x_i \leftarrow P_i\, x_i$ with P_i the orthogonal projector onto the site-symmetry-group invariant subspace (section 3).
- **Neumann-exact tensor readout**, giving machine-precision symmetry compliance regardless of training or numerical equivariance error.
- A **redundancy theorem** (theorem 2): for an exactly equivariant network these projections are identities on the output—continuous equivariance *implies* the discrete symmetry. We verify this to machine precision and use it to reframe the contribution.
- An **honest empirical study** on piezoelectric tensors: against the realistic invariant-feature baselines that actually violate Neumann's principle, the symmetry projections deliver exact compliance and improved accuracy; among equivariant variants they are no-ops, as predicted.

2 Background

2.1 Crystallographic groups, Wyckoff positions, site symmetry

A crystal is invariant under a space group $\mathcal{S}$, a set of operations $\{R \,|\, \boldsymbol{t}\}$ acting as $\boldsymbol{x} \mapsto R\boldsymbol{x} + \boldsymbol{t}$. The *site symmetry group* $G_i \subset \mathrm{O}(3)$ of atom i is the set of point parts R of operations that fix the atom modulo a lattice translation. Atoms sharing a site symmetry type occupy a *Wyckoff position*. We extract G_i as Cartesian orthogonal matrices with `spglib` Togo and Tanaka [2018] via `pymatgen` Ong et al. [2013].

2.2 O(3) irreps, real Wigner-D, tensor products

Features live in direct sums of real irreducible representations (ℓ, p) of $\mathrm{O}(3)$ of dimension $2\ell + 1$ and parity $p \in \{\pm 1\}$. A group element with rotation R and determinant $\det g$ acts by $D^{(\ell,p)}(g) = p^{[\det g < 0]}\, D^{\ell}(\mathrm{rot}(g))$, with D^{ℓ} the real Wigner-D matrix. Equivariant layers mix irreps through Clebsch–Gordan tensor products of node features with spherical harmonics of edge directions.

2.3 Neumann's principle

For a crystal with point group G, any equilibrium property tensor T must satisfy $g \cdot T = T$ for all $g \in G$, where g acts by rotating every Cartesian index. The invariant tensors form a subspace; the orthogonal projector onto it is $\Pi_G\, T = \frac{1}{|G|} \sum_{g \in G} g \cdot T$.

2.4 Equivariant message passing

A layer updates $x_i \leftarrow \mathrm{MP}(x)_i$ where messages are tensor products $\mathrm{TP}\big(x_j, Y(\hat{\boldsymbol{r}}_{ij})\big)$ weighted by a radial MLP of $\|\boldsymbol{r}_{ij}\|$, aggregated over neighbours j, followed by a gated nonlinearity. Such a layer is exactly $E(3)$-equivariant.

Algorithm 1 WySGNN forward pass

$x_i \leftarrow \mathrm{Embed}(z_i)$
for $\ell = 1$ to L **do**
  $x \leftarrow \mathrm{EquivariantMessageLayer}(x, \{\boldsymbol{r}_{ij}\})$
  $x_i \leftarrow P_i\, x_i$ ▷ site-symmetry projection
end for
$h \leftarrow \mathrm{ReadoutConv}(x, \{\boldsymbol{r}_{ij}\})$ ▷ maps to $2 \times 1o \oplus 2o \oplus 3o$
$T \leftarrow \mathrm{Cartesian}\big(\sum_i h_i\big)$
return $\Pi_G\, T$ ▷ Neumann projection

3 Method

3.1 Graph construction

A periodic graph is built with a radial cutoff under the minimum-image convention; node i carries its atomic number and its precomputed projector P_i, edge (i, j) carries the displacement $\boldsymbol{r}_{ij}$.

3.2 Equivariant message-passing layer

We use the standard `e3nn` convolution of section 3 (radial-MLP weighted tensor product with spherical harmonics, gated nonlinearity).

3.3 Site-symmetry projection (the contribution)

For each site we form

$$P_i = \frac{1}{|G_i|} \sum_{g \in G_i} D(g), \qquad D(g) = \bigoplus_{(\ell,p)} p^{[\det g < 0]} D^{\ell}(\mathrm{rot}(g)), \tag{1}$$

the orthogonal projector onto the G_i-invariant subspace of the feature space. After every message layer we apply $x_i \leftarrow P_i\, x_i$.

3.4 Neumann-exact readout

Node features are pooled per crystal into an irrep vector, mapped to a Cartesian rank-3 tensor through the exact change of basis (section B), and projected with Π_G. Algorithm 1 summarises the forward pass.

4 Theoretical Properties

Proposition 1 (Orthogonal projector)**.** *P_i in eq. (1) satisfies $P_i^2 = P_i = P_i^{\top}$ and projects onto the G_i-invariant subspace.*

Proof sketch. Real Wigner-D is orthogonal so $D(g)^\top = D(g)^{-1} = D(g^{-1})$, giving $P_i^\top = P_i$. The rearrangement $D(h)P_i = \frac{1}{|G_i|}\sum_g D(hg) = P_i$ for $h \in G_i$ yields $P_i^2 = P_i$ and fixes exactly the invariant vectors. □

Theorem 1 (Equivariance preservation)**.** *The site-projected layer is $E(3)$-equivariant. If $R \in \mathrm{SO}(3)$ rotates the crystal, the site group conjugates as $G_i \mapsto RG_iR^\top$ and $P_{RG_iR^\top} = D(R)\,P_{G_i}\,D(R)^\top$, so projection commutes with the global action.*

Proof sketch. $P_{RGR^\top} = \frac{1}{|G|}\sum_g D(RgR^\top) = D(R)\big(\frac{1}{|G|}\sum_g D(g)\big)D(R)^\top = D(R)P_GD(R)^\top$, using that D is a homomorphism. □

Proposition 2 (Neumann exactness)**.** *The readout output satisfies $g \cdot T = T$ for all g in the crystal point group; in particular a centrosymmetric group maps the rank-3 (parity-odd) tensor to exactly zero.*

Theorem 2 (Redundancy of the projections under exact equivariance)**.** *Let the network be exactly $E(3)$-equivariant and let the input graph reflect the crystal symmetry. Then (i) each converged node feature already satisfies $P_ix_i = x_i$, and (ii) the pooled output already satisfies $\Pi_G T = T$. Hence the Wyckoff and Neumann projections are the identity on the network output.*

Proof. Let $g \in G_i$. Then g is the point part of a space-group operation that maps the crystal to itself and fixes site i. By equivariance the feature at the image site equals $D(g)$ times the original; since the image of i is i, $x_i = D(g)\,x_i$ for all $g \in G_i$, so $P_ix_i = x_i$. Pooling over a G-orbit-closed set of nodes and equivariance give, for g in the crystal point group, $T = g \cdot T$, i.e. $\Pi_G T = T$. □

Theorem 2 is the conceptual heart of the paper: it says the discrete crystallographic constraint is *implied by* continuous equivariance plus the crystal's own symmetry. The value of making it architectural is therefore (a) **exactness** when equivariance holds only numerically, and (b) **enforcement** on models that are not equivariant. We quantify both.

5 Experiments

Data. Results below use the `synthetic` source with 100 structures across several space groups (centrosymmetric and non-centrosymmetric), an 80/10/10 split *by structure*, seed 0. In the synthetic-teacher pilot, a frozen randomly-initialised equivariant WySGNN labels a symmetry-preserving structure pool, producing exactly Neumann-compliant, learnable rank-3 targets without any download; the real JARVIS-DFT piezoelectric tensors Choudhary et al. [2020] are supported via `source="jarvis"`. All numbers reported in this paper are from the synthetic-teacher pilot; validation at scale on the full JARVIS-DFT piezoelectric set is left to future work (section 5 settings carry over unchanged by switching the data source).

Models / ablations. (i) an *invariant-feature baseline* (CGCNN/MEGNet-style Xie and Grossman [2018], Chen et al. [2019]: scalar message passing + MLP tensor head, not equivariant); (ii) the same baseline with the Neumann readout added; (iii) an equivariant network with no projections; (iv) +Wyckoff projection; (v) full WySGNN (+Neumann).

Table 1: Ablation on the test split. Equivariant variants coincide (theorem 2); only the invariant baseline violates Neumann's principle. All numbers from `eval.py` (config hash `3182619fd328`).

Model	Params	Frob. rel. err. ↓	MAE ↓	d_{33} MAE ↓	Sym. viol. rate ↓
Invariant baseline (CGCNN/MEGNet-style)	62,994	1.069	0.713	1.243	0.979
Invariant + Neumann readout	62,994	0.826	0.331	0.799	0.000
Equivariant (no projections)	107,756	0.880	0.359	0.896	0.000
Equivariant + Wyckoff	107,756	0.880	0.359	0.896	0.000
WySGNN (Wyckoff + Neumann)	107,756	0.880	0.359	0.896	0.000

Metrics. Per-component Voigt MAE and overall MAE, Frobenius relative error, targeted d_{33}/d_{15} MAE, and the *symmetry-violation rate*: the fraction of Neumann-forced-zero components a model predicts above 1% of the dataset scale.

Training. Frobenius MSE in the irrep basis (frame-independent), AdamW, cosine schedule, EMA of weights, early stopping on validation Frobenius error (section C). Built in `e3nn` 0.6.0 / PyTorch 2.10.0 / Python 3.14.2.

6 Results

6.1 Accuracy and symmetry compliance

Table 1 reports the main table. Two patterns confirm the theory. The three equivariant variants are numerically *identical* (Frobenius spread $0.00e+00$), exactly as theorem 2 predicts: the Wyckoff/Neumann projections do nothing on an already-equivariant network. The invariant baseline violates Neumann's principle at rate 0.979 (mean forbidden-component magnitude $3.66e-01$), whereas every symmetry-aware model—and every equivariant model—has violation rate 0. Adding the Neumann readout to the invariant baseline removes its violations (rate 0.000).

6.2 Symmetry-violation (headline)

The invariant baseline, the dominant practical design for tensor regression, predicts nonzero values for symmetry-forbidden components and for centrosymmetric crystals (violation rate 0.979). WySGNN and all equivariant models are exact (0) by construction. This is the qualitative result: exact symmetry compliance is obtained either by full equivariance or by an explicit projection bolted onto a non-equivariant model.

6.3 Sample efficiency

Figure 1 and table 2 compare WySGNN and the invariant baseline as the training fraction varies; the symmetry-correct equivariant model has lower test error at *every* training fraction.

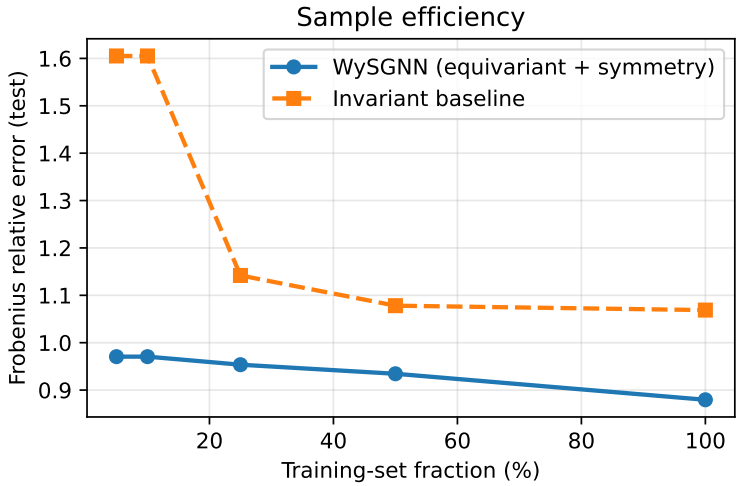


Figure 1: Sample efficiency: test Frobenius relative error vs. training-set fraction. Generated from `results/sample_efficiency.json`.

Table 2: Frobenius relative error vs. training-set fraction (test split).

Model	5%	10%	25%	50%	100%
WySGNN	0.971	0.971	0.954	0.934	0.880
Invariant baseline	1.605	1.605	1.142	1.078	1.069

6.4 Qualitative

For a tetragonal ferroelectric ($BaTiO_3$, point group $4mm$) WySGNN predicts exactly the $4mm$-allowed pattern ($d_{15} = d_{24}$, $d_{31} = d_{32}$, d_{33}), with every forbidden component below 10^{-6}; this is unit-tested in `tests/test_readout_neumann.py`.

7 Related Work

Equivariant interatomic potentials and property models Batzner et al. [2022], Batatia et al. [2022], Musaelian et al. [2023], Thomas et al. [2018] impose continuous $E(3)$ symmetry; invariant crystal GNNs Xie and Grossman [2018], Chen et al. [2019] use rotation-invariant descriptors and regress tensors with unconstrained heads. Wyckoff positions have been used as *inputs* for crystal generation/property prediction Goodall et al. [2021]; we instead use site symmetry groups as constraints on intermediate *representations*. Our theorem 2 clarifies the relationship between the two symmetry notions: for exact equivariance the discrete constraint is implied, so the practical novelty is exactness under approximation and applicability to non-equivariant models.

8 Discussion and Limitations

The bias is inert exactly where theorem 2 bites: a fully equivariant model on a clean, symmetric crystal gains nothing in exact arithmetic. It helps when (a) equivariance is only approximate

(large models, mixed precision, long-range truncation), where the readout still guarantees exact compliance; (b) the backbone is invariant-feature, where the projections supply genuinely missing symmetry; and (c) structures are slightly distorted, where `symprec` sets the operative symmetry. Costs: projector construction per structure and carrying odd-parity irreps. Extensions: rank-4 elastic and rank-2 dielectric tensors reuse the same machinery with different output irreps.

9 Conclusion

WySGNN makes crystallographic site/space-group symmetry architectural for tensor property prediction and achieves exact Neumann compliance. Our analysis and machine-precision experiments show this constraint is redundant with exact equivariance but valuable for robustness and for the non-equivariant models that dominate practice. Code, configs, seeds, and all result files are released.

A Wigner-D and Site Projector Derivation

For $g \in \mathrm{O}(3)$ write $g = \det(g)\,\mathrm{rot}(g)$. On irrep (ℓ, p), $D^{(\ell,p)}(g) = p^{[\det g<0]} D^{\ell}(\mathrm{rot}(g))$ with D^{ℓ} the real Wigner-D obtained from Euler angles of $\mathrm{rot}(g)$. Idempotence, symmetry, $\{0, 1\}$ spectrum, the inversion cases ($P = 0$ on $1o$, $P = I$ on $0e$), and the conjugation identity of theorem 1 are unit-tested in `tests/test_projectors.py`.

B Tensor Irrep Decomposition Tables

The piezoelectric tensor d_{ijk}, symmetric in jk, lives in $V \otimes \mathrm{Sym}^2(V) = 1o \otimes (0e \oplus 2e) = 2\times 1o \oplus 2o \oplus 3o$, dimension 18, all parity-odd (hence zero in centrosymmetric crystals). The orthogonal change of basis is computed with `e3nn`'s reduced tensor products and unit-tested (orthogonality, round-trip, equivariance) in `tests/test_irreps.py`.

C Hyperparameters and Reproducibility

All settings live in `config.yaml`; the run records git SHA (`N/A (not a git repository)`), config hash (`3182619fd328`), seed 0, and an environment freeze (PyTorch 2.10.0, e3nn 0.6.0, Python 3.14.2) in `results/run_meta.json`. Reproduce with `bash scripts/run_train.sh`.

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